

Semantic Communication for Edge AI

Task-oriented deep joint source–channel coding for image classification

This project compares three ways to transmit an image for remote classification over a bandwidth-constrained noisy channel.

Capstone First Review

Review window: 18–22 August 2026

Presentation slot: 15 minutes

Team / guide details to be added

Project context and basic definition

The application is remote image classification. An edge device observes an image and sends information across a limited, noisy link to a receiver that owns the classifier.

Semantic communication

Semantic communication transmits the information required to preserve meaning or complete a downstream task. It does not require every original source bit to be reconstructed when the task does not need them.

- Source: an image observed by an edge camera or sensor.
- Channel constraint: fixed bandwidth and additive wireless noise.
- Receiver task: predict the correct image class.

Tier 1 is simulation-first. SDR and edge-device deployment are optional later stages.

Problem statement and research question

Conventional approach

A conventional link compresses an image, protects the bits with a channel code, reconstructs the image at the receiver, and then runs the classifier. Its interfaces optimize image fidelity and bit recovery.

Task-oriented approach

A learned joint system trains the sender and receiver together through a differentiable channel. Its representation can be optimized directly for classification after channel noise.

Research question

At the same bandwidth and channel conditions, can task-aware joint coding preserve classification accuracy better than a properly tuned separated system?

The project reports the measured result even if the learned system does not outperform the baseline.

Related work and the gap addressed

Finite-blocklength communication	Short packets operate below asymptotic capacity and incur non-negligible coding overhead and error probability. [1–4]
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Learned image compression	End-to-end transforms improve rate–distortion performance, but reconstruction quality is not the same as task accuracy. [11–17]
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Deep joint source–channel coding	DeepJSCC studies continuous learned mappings and graceful degradation over noisy channels. Most image results emphasize reconstruction. [5–10,24]
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Task-oriented communication	Learned representations can preserve inference performance, but task awareness and joint coding must be separated experimentally. [18–23]
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Gap addressed	A three-way comparison at equal channel uses: adaptive classical image transmission, task-aware digital features, and learned joint coding.
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Project objectives and completion criterion

1. Implement a normalized simulated AWGN channel and a task-oriented DJSCC system.
2. Implement a JPEG 2000 + 5G NR LDPC baseline that is tuned on validation data.
3. Implement a task-aware digital feature system using the same digital physical layer.
4. Match channel uses, SNR definition, image identity and noise identity across systems.
5. Freeze all choices before one paired evaluation on the sealed test split.
6. Report positive, null or negative findings using the preregistered analysis.

Completion criterion

Tier 1 is complete when all three systems are implemented, fairly matched, evaluated once, and reported. Completion does not require a positive result.

Systems under comparison

System	Sender and channel path	Receiver output
Classical image link	JPEG 2000 → packet framing → 5G NR LDPC → modulation → AWGN	Reconstructed image → frozen classifier
Digital feature link	Learned features → quantization → same LDPC and modulation → AWGN	Recovered features → task head
Learned joint link	Neural encoder → power normalization → AWGN	Dual-head neural decoder → class

Shared inputs

Same image, dataset split, complex-symbol budget, SNR definition and keyed channel realization.

Data isolation

Validation is used for selection. The test split remains sealed until gate G-12.

Experimental controls and fairness rules

Control	Rule
Resource matching	Compare systems at the same complex-symbol budget and SNR definition.
Classical baseline strength	Tune JPEG 2000 quality, LDPC rate and BPSK/QPSK/16-QAM per SNR on validation.
Complete accounting	Count container, CRC, segmentation and filler overhead. Keep decode failures in the denominator.
BLER evidence	Measure every required physical-layer identity. Missing evidence is not interpreted as zero error.
Attribution	Use the digital feature system to separate task-aware representation from joint coding.
Evaluation	Use paired per-image outcomes and one sealed test campaign after all choices are frozen.

Engineering and gate evidence—not learned-vs-classical results

Gate	Status	Recorded evidence	Meaning
G-1	PASS	898/1,000 = 89.8% validation accuracy	The reference classification task is viable.
G-7	PASS	1.64 M parameters; 48.68 s per epoch; 1.004 GiB VRAM	The planned DJSCC model fits the available GPU.
G-2	PASS	Golden vectors and BLER waterfall checks within 0.5 dB	The digital physical layer passes conformance checks.
W4	PASS	Bounded JPEG 2000 → LDPC → AWGN → decode run	The classical pipeline and evidence path work end to end.
G8_C	ACTIVE	748 / 3,213 authenticated BLER units	The full BLER table and operating-point selection are not complete.

No learned-versus-classical headline result exists yet. Bounded integration evidence is not presented as a scientific comparison.

Remaining work and review dates

Period / gate	State	Work	Output
Now · G8_C	Active	Complete BLER characterization and freeze the table	Inputs for G-8
G-8	Planned	Select operating ratios and classical configurations	Frozen baseline choices
W5–W8	Planned	Train DJSCC; calibrate λ ; implement digital feature control	Frozen learned systems
W9–W11	Planned	Validation rehearsal; freeze; one test campaign	Hypothesis decisions
W13–W17	Planned	Demo, report, poster and final review preparation	Final package

First Review: 18–22 Aug · Second Review: 29 Sep–3 Oct · Final Review: 17–21 Nov · Report due: 20 Nov

W16 is reserved for report contingency. It is not available for new experiment scope.

Scope, deliverables, and decisions requested

Required Tier 1 scope	Simulation-based comparison of the three systems on image classification over normalized AWGN, with one sealed test campaign and paired analysis.
Optional later scope	Conducted SDR replay and Raspberry Pi deployment are stretch goals after gate G-5. No hardware has been purchased.
Technical claim boundary	OpenJPEG 2.5.4 and a TS 38.212-derived LDPC/rate-matching chain over abstract AWGN. This is not a full 5G NR link claim.
Decision requested	Confirm the task boundary, the three-system comparison, the fairness controls, and simulation-first Tier 1 as a sufficient completion path.
Outstanding human item	The guide's dated acknowledgement of the simulation-first, no-required-hardware path is still pending.

Hypotheses and analysis plan

H1 — Low-SNR separation (confirmatory primary)

Comparator: learned versus classical_adaptive at the headline ratio. A grid point qualifies when the studentized paired mean $\sqrt{N} \cdot \Delta(\hat{s}) / \sigma(\hat{s})$ exceeds 1.96. Let R_{obs} be the longest run of consecutive qualifying points at or below the training SNR. H1 is supported only if $R_{\text{obs}} \geq 3$ and the calibrated run p-value ≤ 0.05 .

H2 — Graceful versus cliff

The fixed classical curve must show the preregistered large drop while the learned curve remains within its smaller-drop bound.

H3 — Convergence

The paired learned-minus-classical gap must contract across SNR. A crossover is reported if observed but is not required.

H4 — Attribution

Joint coding is credited only if the learned joint system also exceeds the task-aware digital feature control under the H1 procedure.

All hypotheses use paired per-image outcomes. The test split is opened once after gate G-12.

First Review rubric map and sources

Criterion	Slides	Evidence	Selected references
Motivation	2–3	Application context and problem statement	[1] Shannon, 1948. [3] Polyanskiy, Poor & Verdú, 2010.
Objectives	3, 5–7, 10	Systems, completion criterion and review ask	[4] Kostina & Verdú, 2013.
Hypothesis	5, 11	Outcome-independent completion and formal rules	[5] Bourtsoulatze, Kurka & Gündüz, 2019.
Problem Survey	4, 12	30-source literature synthesis	[11] Ballé, Laparra & Simoncelli, 2017.
Subject Knowledge	6–8, 10–11	System design, controls, evidence and scope	[17] Blau & Michaeli, 2018.
Time Plan	9	Gate-ordered schedule and fixed review dates	[18] Jankowski, Gündüz & Mikołajczyk, 2021. [20] Xie et al., 2021.
Evidence records			Refresh G8_C status from instructions/RESUME.md on submission day.
G-1: results/reference_classifier/g1_adjudication.json G-7: results/profiling/g7_djscc_profile.json G-2 / W4 / G8_C: results/baseline/			